

Final Exam

Time Limit: 120 minutes

- Every non-trivial calculation and claim in the exam must be properly justified.
- Your work must be well organized and coherent. Pay attention to notation, definitions of events, units of measurement, and interpretation.
- You must answer each question inside the designated box; answers outside will not be considered.
- The use of calculators is allowed.
- Digital devices such as mobile phones, tablets, and laptops, as well as internet access, are forbidden.

The following questions are about the housing affordability crisis in a fictional city, *Puerto Clara*. The city is considering housing vouchers, rent-stabilization rules, and targeted subsidies to reduce rent burden and displacement. The scenario is inspired by empirical research on housing voucher lotteries and rent-control policies, but all numerical values in this exam are constructed for the exam.¹

¹For background inspiration only: Jacob, Kapustin, and Ludwig (2014), “Human Capital Effects of Anti-Poverty Programs: Evidence from a Randomized Housing Voucher Lottery”; Diamond, McQuade, and Qian (2019), “The Effects of Rent Control Expansion on Tenants, Landlords, and Inequality: Evidence from San Francisco”.

1. (2.5 points) The Housing Office audits 600 rental applications. For each applicant, it records whether the household has children and whether the household is *rent-burdened*, defined as spending more than 40% of monthly income on rent.

	Rent-burdened (B)	Not rent-burdened (B^c)
Family with children (C)	168	112
No children (C^c)	132	188

- (a) Compute $P(C)$, $P(B)$, $P(C \cap B^c)$, and $P(B | C)$.

- (b) What does it mean that the events C and B are not independent? Explain the intuition behind the definition in words.

- (c) The Housing Office also uses an algorithm that flags a household as being at high risk of housing insecurity (event H). Suppose the algorithm flags 80% of rent-burdened households and 20% of households that are not rent-burdened. Compute $P(B | H)$. Does knowing that a household was flagged change your mind about how likely it is for the household to be rent-burdened? If so, quantify this change of belief.

- (d) Let $I_i = 1$ if applicant i is rent-burdened and $I_i = 0$ otherwise. Suppose future applicants are independently rent-burdened with probability $P(B) = 0.5$. For a random sample of $n = 100$ future applicants, let

$$\hat{p} = \frac{1}{100} \sum_{i=1}^{100} I_i$$

be the sample proportion of rent-burdened applicants.

- (i) What does the Law of Large Numbers say about $\hat{p}$ as the sample size becomes large?

- (ii) What is the distribution of $100\hat{p} = \sum_{i=1}^{100} I_i$?

2. (2.5 points) In 2024, Puerto Clara introduced a rent assistance program in its East district. The West district did not receive the program. The outcome is the average number of formal eviction cases started by landlords per 1,000 renter households.

District	2023 cases	2024 cases
East district	18	13
West district	12	10

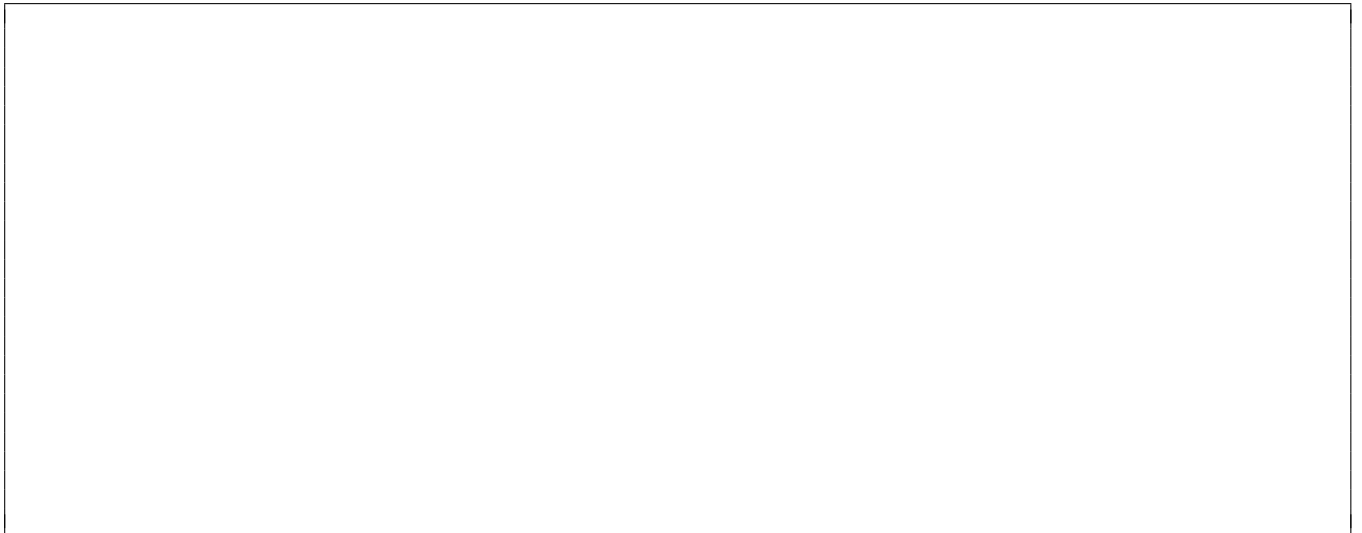
- (a) Identify the treatment variable, outcome variable, treatment group, control group, before period, and after period.

- (b) Compute the Before-and-After estimator for the East district. Would you rely on this as a causal estimate? Explain.

- (c) Compute the Difference-in-Differences estimator.



- (d) What is the key assumption behind the Difference-in-Differences estimator? Give one example of a violation in this context.



3. (5 points) The city runs a randomized lottery among eligible low-income households, such that all lottery winners receive and use a housing voucher. One year later, the city estimates the following regression:

$$\widehat{housing_quality} = 42.0 + 5.5 income + 9.0 voucher - 0.5 (voucher \times income), \quad R^2 = 0.23.$$

The observed range of *income* in the estimation sample is from 0.8 to 5 thousand euros. The variables are:

- *housing_quality* is a housing quality index, with higher values meaning better housing conditions;
 - *income* is monthly household income, measured in thousands of euros;
 - *voucher* = 1 if the household received a voucher and 0 otherwise.
- (a) Interpret the coefficient on *income* for households without a voucher. How does receiving a voucher change the relationship between income and housing quality?

- (b) Can the coefficient on *voucher* be interpreted as an unbiased average treatment effect? Why or why not?
Hint: compare two households with the same income, one with a voucher and one without.

- (c) Comment on the predictive power of the regression. Would you use it to predict for a household with 10 thousand euros of monthly income? Why or why not?

The city now decides to issue the voucher to all households below 500 euros of monthly income. After gathering new data, the fitted regression is

$$\widehat{housing_quality} = 44 + 5\,income + 8\,voucher - 0.5\,(voucher \times income).$$

- (d) Compute the Regression Discontinuity estimate at the cutoff and interpret it.

- (e) State the key assumption needed for a causal interpretation.